

Lab 7. Feign and Registry

Part 1: Feign

Write 2 services:

1. A StockService that returns how many products we have in stock given a certain productNumber. The number that is returned can be hard coded. This service runs on port 8900.
2. A ProductService that returns a Product given a certain productNumber. The Product contains the fields: productNumber, name, number on stock
The number on stock is retrieved from the StockService. This service runs on port 8901.

Use Feign so that the ProductService can call the StockService.

The ProductService needs the Feign dependency:

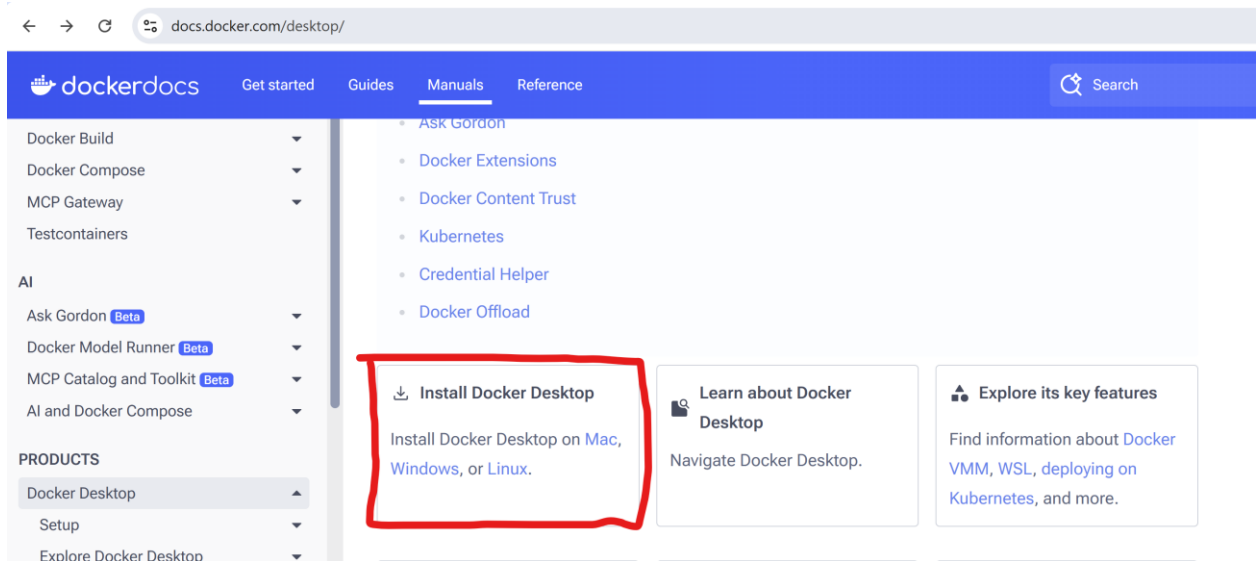
```
<dependency>  
  <groupId>org.springframework.cloud</groupId>  
  <artifactId>spring-cloud-starter-openfeign</artifactId>  
</dependency>
```



Part 2

Installing the consul registry:

First go to <https://docs.docker.com/desktop/>



Then choose your operating system. (I choose Windows)

[Home](#) / [Manuals](#) / [Docker Desktop](#) / [Setup](#) / [Install](#) / [Windows](#)

Install Docker Desktop on Windows

Page options ▼

Docker Desktop terms

Commercial use of Docker Desktop in larger enterprises (more than 250 employees OR more than \$10 million USD in annual revenue) requires a [paid subscription](#).

This page provides download links, system requirements, and step-by-step installation instructions for Docker Desktop on Windows.



[Docker Desktop for Windows - x86_64](#)

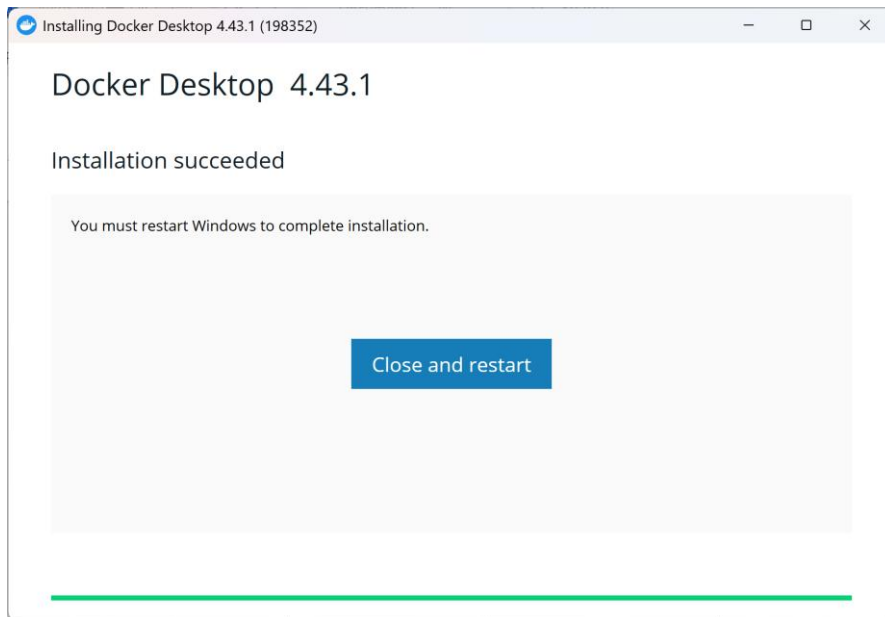
[Docker Desktop for Windows - x86_64 on the Microsoft Store](#)

[Docker Desktop for Windows - Arm \(Early Access\)](#)

For checksums, see [Release notes](#)

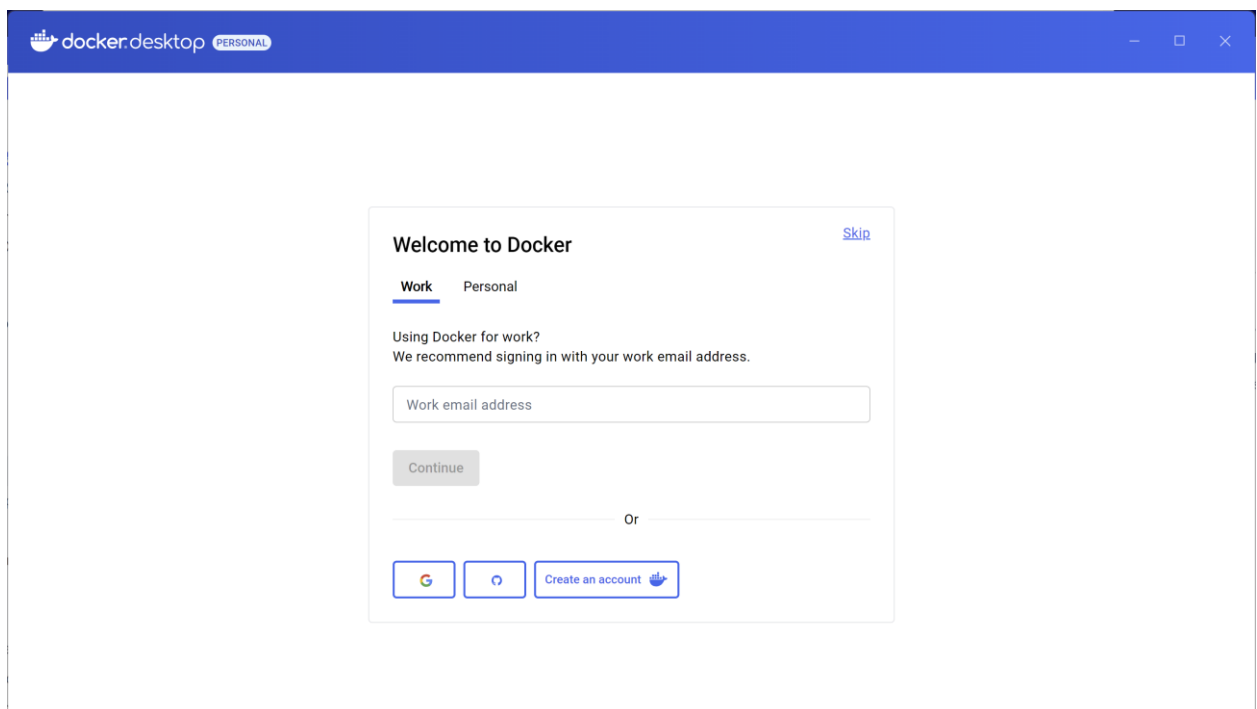
Click the button to download Docker Desktop Installer.

Then run the Docker Desktop Installer till installation has succeeded.

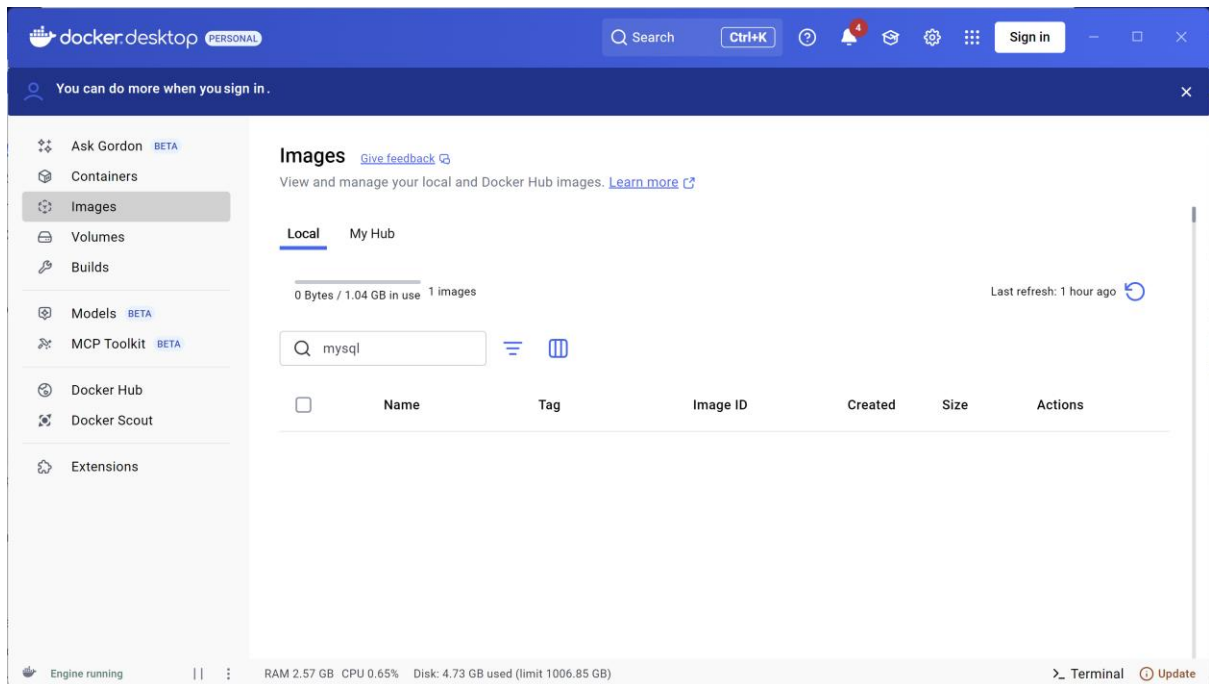


Click the **Close and restart** button.

Now you should be able to run Docker Desktop



Click the **Skip** link.



Now open a prompt window and check if docker is installed:

```

Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ictin>docker -v
Docker version 28.3.2, build 578ccf6

C:\Users\ictin>

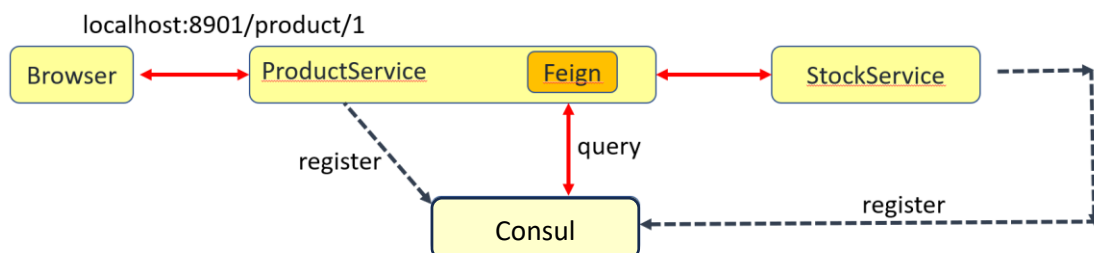
```

Now we are going to install and start consul.

In the command prompt, type the following command:

docker run -d -p 8500:8500 --name=consul hashicorp/consul:latest

Then if you call the ProductService, it should call to the StockService using the registry.



Then configure the ProductService and the StockService so that they use the Consul registry.

Run the StockService and see in the Consul dashboard if it is registered.

Run the ProductService and see in the Consul dashboard if it is registered.

Then if you call the ProductService, it should call to the StockService using the registry.

Part 3

In Lab 4 part 2 we implemented a modular monolith with a Product component and a shoppingcart component. Take the solution of lab 4 part 2 and modify it to a microservice architecture. This means we get a separate project for the product service and a separate project for the shoppingcart service. Then we can use the client project that call these separate components.

Make sure that when we add a product to the shopping service, that the shopping service calls the product service to get product data. (Make sure the components use the registry to call each other)

Also the client will update the price of a product in the product service. Make sure that the corresponding products in the shoppingcars in the shoppingservice will also be updated.

What to hand in?

1. A zip file containing all service projects for part 1
2. A zip file containing all service projects for part 2
3. A zip file containing all service projects for part 3